
INSTRUCTIONS TO THE STUDENTS:-

- 1) Question 1 is compulsory and carries 10 marks whereas Question no 2 to Question no 7 carries 12 marks each.
 - 2) Attempt any 5 Questions from Question 2 to Question 7
 - 3) Illustrate the answers with neat sketches, diagram etc. wherever necessary.
 - 4) Necessary data is given in the respective questions. If such data is not given it means its knowledge is a part of examination.
 - 5) If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.
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Q.1 Attempt the following

- I) Which of the following is a type of line configuration.
A. Multi Point
B. Ring
C. Full duplex
D. None of these
- II) RARP converts
A. Logical to Physical Address
B. Physical to Logical Address
C. Physical to Port Address
D. IP to Physical Address
- III) Message Authentication can be done by
A. MDC
B. Message Digest
C. Digital Signature
D. Password
- IV) ICMP packet is used for
A. Control message
B. Error Reporting
C. Query Message
D. All of above
- V) Vulnerable time for Slotted ALOHA is
A. T_{fr}
B. $2 T_{fr}$
C. $T_{fr} + T_p$
D. T_p
- VI) To avoid the collision using Closs criteria, the total number of cross points required for three stage cross bar switch with 200 inputs, $n=10$ & $K=2$ is
- VII) Find the value for checksum of 0x4A, 0x5B, 0x6C, 0x7D.
- VIII) For hamming codes, the value of k & n to send a dataword of 16 bit is
- IX) The Netid & Hostid for IP address 193.147.200.255 is
- X) Find the Cipher Text equivalent for Plain Text **ELECTRONIC** using cipher as,

Plain Text Position 1 2 3 4 5

Cipher Text Position 4 3 5 2 1

PTO

Q.2

- I) Match the following with one or more layers of OSI according to their function 8M
- A. Finalize path for packets
 - B. Responsibility for carrying frames between adjacent nodes.
 - C. Access to network resources
 - D. Line configuration
 - E. Delivery of entire message
 - F. Map logical address to physical address
 - G. Synchronization of data
 - H. Type of connection
- II) What is the value of bit rate and signal level required for 1 MHz channel with 63 SNR. 4M

Q.3

- I) Draw Differential Manchester, 2B1Q, MLT-3, HDB-3 line codes for 0001100001010101. 8M
- II) Write a short note on(Any One) 4M
- A. Virtual Circuit Network
 - B. Unguided Transmission Media

Q.4

- I) Write the simple parity check codeword for dataword 1010. 2M
- II) Find the CRC (7,4) codeword for sending dataword 1001 with generator polynomial as x^3+x+1 . 5M
- III) Explain in brief how minimum hamming distance is used for error detection & correction. 5M

Q.5

- I) Draw the timing diagram using Go Back n-ARQ protocol having 3 bit sequence numbers & show the transmission for 8M
- A. Frame 0 sent without any loss.
 - B. Frame 1, 2 transmitted & lost during transmission.
 - C. Frame 3 is transmitted & received at receiver.
 - D. Complete the further till window slides by atleast one slot.
- II) Explain the following in detail (Any one) 4M
- A. Controlled Access
 - B. CSMA/CA

Q.6

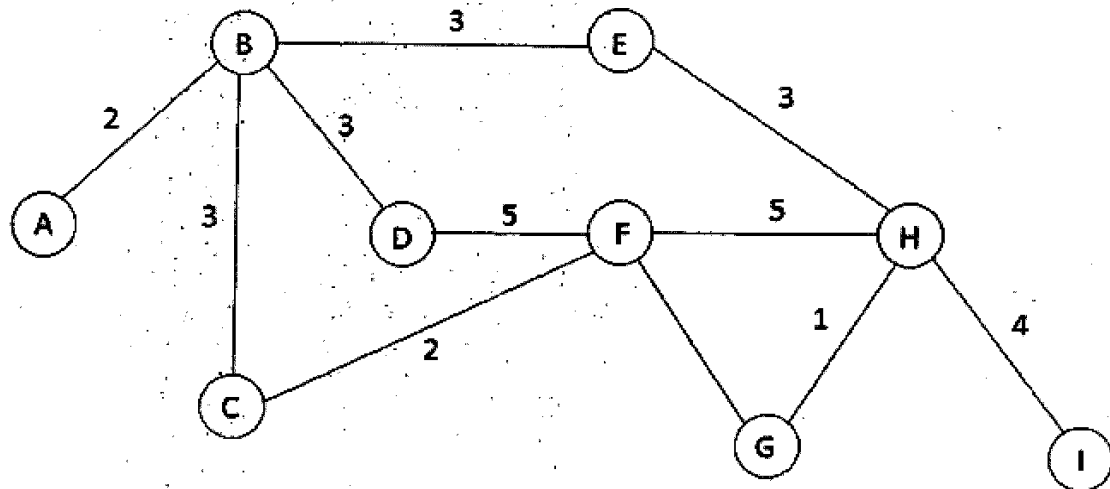
- I) An ISP is granted a block of addresses starting with 200.200.0.0/14. The ISP needs to distribute these addresses to three groups of customers as follows: 6M
- A. The first group has 256 customers; each needs 128 addresses.
 - B. The Second group has 128 customers; each needs 256 addresses.
 - C. The third group has 128 customers; each needs 64 addresses.

Design the sub blocks and give slash notations for each sub block. Find out how many addresses are still available after these allocations.

II) Explain in brief TCP Connection. 6M

Q.7

I) Design a routing table at each node for the following network: 8M



II) Write a short note on: (Any One) 4M

- A. Diffie-Hellman Algorithm
- B. Message confidentiality using cryptography

*** BEST OF LUCK ***